

IR Assignment 1: Inverted Index & Vector Space Model

DEPARTMENT OF COMPUTER ENGINEERING - ASSIGNMENT 1

Subject: Information Retrieval | Submission Deadline: Week 4

Task Overview:

1. Implement a text preprocessing pipeline consisting of tokenization, lowercase conversion, and stop-word elimination.
2. Construct an Inverted Index mapping distinct vocabulary terms to document IDs.
3. Compute TF-IDF matrices and implement Cosine Similarity retrieval ranking for multi-word queries.
4. Evaluate the retrieved document rankings using Precision, Recall, and F1 score against a ground-truth relevance test set.